PUPI - YOUR LITTLE AI PET COMPANION
Virtual Pet AI Challenge - Final Master Development & Submission Document

1. PRODUCT IDENTITY & CORE ARCHITECTURAL PRINCIPLE
Product: PUPI - Your Little AI Pet Companion
Platform: Android smartphones only.
Core concept: A small interactive 3D puppy that floats above supported Android applications.
Pupi lives natively on the screen during everyday smartphone use.

STRICT PRODUCT DECISION:
- There is NO dedicated Pupi pet-care application experience.
- No separate pet-care app or pet home screen.
- The floating 3D puppy is the entire product experience.
- Minimal Android host interface is permitted solely as an onboarding gate for SYSTEM_ALERT_WINDOW permission, battery optimization whitelisting, and microphone access.

2. RECOMMENDED ANDROID TECHNOLOGY \(Optimized for Intel i3-1215U / 8GB RAM\)
3D Rendering Engine: Google Filament \(PBR\)
Justification:
- Lightweight C++ core with Android Kotlin bindings \(only ~6MB native libs\).
- Compiles rapidly without the 4GB+ RAM penalty and slow editor launches of Unity.
- Provides Physically Based Rendering \(PBR\), transparent TextureView overlays, and skeletal animation.
- Dynamic frame throttling: 0 FPS when stationary idle, 60 FPS during touch and movement.
Window Overlay: Android WindowManager \(TYPE_APPLICATION_OVERLAY\) bound to a Foreground Service.
State & Persistence: Room Database \(SQLite + Flow\) for offline-first reactive persistence.
Voice & Audio: Native Android SpeechRecognizer and TextToSpeech \(zero cloud dependencies\).

3. CHARACTER DESIGN - PUPI
Visual Appearance:
- Soft white body with warm caramel-brown patches over the left eye and lower back.
- Large expressive warm amber/brown eyes with procedural specular highlight.
- Floppy brown ears with natural secondary physics bounce.
- Small velvety puppy nose and rounded snout with a gentle smile crease.
- Bright red nylon collar with a yellow brass tag engraved with PUPI.
- Proportions: 4,800 polygons, 32 bones, optimized for smooth 60 FPS mobile rendering.
Personality: Curious, playful, mischievous, innocent, friendly, observant, kind-hearted.

4. TASK 1 - THE STORY OF PUPI: A HEARTBEAT ON THE GLASS
The first time Julian saw Pupi, it was 11:42 PM on a Thursday. Julian’s phone screen was buried beneath a spread of half-read PDF lecture notes and a cold cup of tea. He clicked the power button, bracing for the usual flood of stressful group-chat pings and deadlines.

Instead, the screen lit up with a tiny soft white puppy sitting quietly just above his dock icons. Pupi had floppy caramel-brown ears, a bright red collar, and a yellow medallion that glinted softly as he lifted his head. When Julian stared in surprise, Pupi blinked twice, tilted his head with an inquisitive soft sound, and took three hesitant steps before rolling onto his belly.

Julian reached out and tapped the puppy’s head. Pupi sprang up, eyes sparkling, tail wagging

in a joyous blur as he nuzzled right against the point where Julian's finger met the glass.

In the days that followed, Pupi became an inseparable presence. He lived with Julian across his everyday phone usage. When Julian scrolled news feeds, Pupi trotted along the bottom edge, curiously peering at headlines. When Julian spent two uninterrupted hours doom-scrolling, Pupi yawned conspicuously, curled into a ball on the screen boundary, and fell asleep, softly reminding Julian how fast time was slipping away.

When Julian had to study, Pupi held up a tiny water droplet icon, gently barking with a speech reminder: 'Pupi thinks you should take a sip of water, Julian!' Julian would laugh, drink water, and watch Pupi do a victorious celebratory bounce. He wasn't an app to open and close; he was a loyal companion living in the margins of digital noise, bringing warmth to every unlock.

5. TASK 2 - FOUR-EMOTION ILLUSTRATION SPECIFICATION
Requirements: Clean black-and-white line art, pure white background, 4 distinct expressions:
1. HAPPY: Crescent closed eyes, perky upright floppy ears, wide joyful smile, radiant posture.
2. SAD: Drooping long ears, large downcast watery pupils, small quivering mouth, single teardrop.
3. ANGRY: Furrowed brow, tense sideways ears, adorable pouty grump expression, folded front paws.
4. SURPRISED: Huge wide circular open eyes, ears perked high, exclamation marks, open 'O' mouth.
All panels preserve identical character identity, proportions, and the PUPI collar tag.

6. TASK 3 - MULTI-PHONE ONLINE ARCHITECTURE MAP
Scenario: Phone A feeds Pupi -> Backend validates & mutates state -> Phone B receives real-time update.
Components:
- Android Phone A: Optimistic UI, local Room DB, Pending Action Queue with client UUIDs.
- Android Phone B: Subscribed via WebSocket, receives state updates, triggers reactive animation.
- Backend API Gateway: Authenticates devices \(JWT Auth\), validates action idempotency.
- Pet Management Service: Implements Optimistic Concurrency Control \(OCC\) with atomic version checks.
- Cloud Database \(PostgreSQL / Supabase\): Stores pet state \(hunger, happiness, version\) & action logs.
- WebSocket / MQTT Fanout Broker: Dispatches broadcast updates to all paired family devices.

Solutions to Technical Challenges:
- Simultaneous Actions: Solved via OCC version checks \(WHERE id = ? AND version = ?\).
- Offline Usage: Solved via Local Pending Queue in Room; synced on reconnection via WorkManager.
- Duplicate Requests: Solved via Client Action UUID idempotency checks on backend.
- Security & Privacy: Solved via Row-Level Security \(RLS\) restricting pairing to owner groups.

7. AI TOOLS & HONEST PROMPT EXPERIMENTATION LOG
- Tool: Gemini / Claude for Architecture and Behaviour State Machine design.
  Initial prompt asked for general AI virtual pet; refined to deterministic Finite State Machine in pure Kotlin with discrete stat decay rules. Resulted in robust PupiBehaviorEngine.kt.
- Tool: Imagen 3 for Character Expression line-art generation.
  Initial prompts yielded colored 3D models with inconsistent dog breeds. Refined with explicit negative constraints \('zero color, black and white line art only, identical collar and tag'\).

8. TESTING & COMPLIANCE SUMMARY
- Functional Tests: Overlay permission, touch dragging, stat decay, and persistent Room DB.
- Android Reliability: Handled background limits, screen rotation, and memory constraints.

- Status: Local Floating Companion & Architecture fully verified; Cloud Sync mapped and proposed.